

NPN
MPS6512
 thru
MPS6515
PNP
MPS6516
 thru
MPS6519

CASE 29-02, STYLE 1
TO-92 (TO-226AA)

AMPLIFIER TRANSISTOR

Refer to 2N4125 for graphs.

MAXIMUM RATINGS

Rating	Symbol	NPN	PNP	Unit
Collector-Emitter Voltage MPS6512, MPS6513 MPS6514, MPS6515 MPS6516 thru MPS6518 MPS6519	V_{CE0}	30 25 — —	— — 40 25	Vdc
Collector-Base Voltage MPS6512 thru MPS6515 MPS6516 thru MPS6518 MPS6519	V_{CBO}	40 — —	— 40 25	Vdc
Emitter-Base Voltage	V_{EBO}	4.0	4.0	Vdc
Collector Current — Continuous	I_C	100	100	mAdc
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	625 5.0		mW mW/ $^\circ\text{C}$
Total Device Dissipation @ $T_C = 25^\circ\text{C}$ Derate above 25°C	P_D	1.5 12		Watts mW/ $^\circ\text{C}$
Operating and Storage Junction Temperature Range	T_J, T_{stg}	-55 to +150		$^\circ\text{C}$

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Case	$R_{\theta JC}$	83.3	$^\circ\text{C/W}$
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	200	$^\circ\text{C/W}$

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted.)

Characteristic	Symbol	Min	Typ	Max	Unit
OFF CHARACTERISTICS					
Collector-Emitter Breakdown Voltage ($I_C = 0.5 \text{ mAdc}, I_B = 0$)	$V_{(BR)CEO}$	30 25	— —	— —	Vdc
($I_C = 0.5 \text{ mAdc}, I_B = 0$)		40 25	— —	— —	
Emitter-Base Breakdown Voltage ($I_E = 10 \mu\text{Adc}, I_C = 0$) ($I_E = 10 \mu\text{Adc}, I_C = 0$)	$V_{(BR)EBO}$	4.0 4.0	— —	— —	Vdc
Collector Cutoff Current ($V_{CB} = 30 \text{ Vdc}, I_E = 0$) ($V_{CB} = 30 \text{ Vdc}, I_E = 0$) ($V_{CB} = 20 \text{ Vdc}, I_E = 0$)	I_{CBO}	— — —	— — —	0.05 0.05 0.05	μAdc

ON CHARACTERISTICS

DC Current Gain ($I_C = 2.0 \text{ mAdc}, V_{CE} = 10 \text{ Vdc}$)	MPS6512 MPS6513 MPS6514 MPS6515	h_{FE}	50 90 150 250	— — — —	100 180 300 500	—
($I_C = 100 \text{ mAdc}, V_{CE} = 10 \text{ Vdc}$)(1)	MPS6512 MPS6513 MPS6514 MPS6515		30 60 90 150	— — — —	— — — —	
($I_C = 2.0 \text{ mAdc}, V_{CE} = 10 \text{ Vdc}$)	MPS6516 MPS6517 MPS6518 MPS6519		50 90 150 250	— — — —	100 180 300 500	
($I_C = 100 \text{ mAdc}, V_{CE} = 10 \text{ Vdc}$)(1)	MPS6516 MPS6517 MPS6518 MPS6519		30 60 90 150	— — — —	— — — —	
Collector-Emitter Saturation Voltage ($I_C = 50 \text{ mAdc}, I_B = 5.0 \text{ mAdc}$) ($I_C = 50 \text{ mAdc}, I_B = 5.0 \text{ mAdc}$)		$V_{CE(sat)}$	— —	— —	0.5 0.5	Vdc

SMALL-SIGNAL CHARACTERISTICS

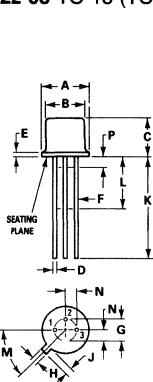
Output Capacitance ($V_{CB} = 10 \text{ Vdc}, I_E = 0, f = 100 \text{ kHz}$) ($V_{CB} = 10 \text{ Vdc}, I_E = 0, f = 100 \text{ kHz}$)	C_{obo}	— —	— —	3.5 4.0	pF
---	-----------	--------	--------	------------	----

(1) Pulse Test: Pulse Width $\leq 300 \mu\text{s}$, Duty Cycle $\leq 2.0\%$.

Package Outline Dimensions

Dimensions are in inches unless otherwise noted.

CASE 22-03 TO-18 (TO-206AA) METAL



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	5.31	5.84	0.209	0.230
B	4.52	4.95	0.178	0.195
C	4.32	5.33	0.170	0.210
D	0.406	0.533	0.016	0.021
E	0.406	0.762	—	0.030
F	0.406	0.483	0.016	0.019
G	2.54 BSC		0.100 BSC	
H	0.914	1.17	0.036	0.046
J	0.711	1.22	0.028	0.048
K	12.70	—	0.500	—
L	6.35	—	0.250	—
M	45° BSC		45° BSC	
N	1.27 BSC		0.050 BSC	
P	—	1.27	—	0.050

All JEDEC notes and dimensions apply.

CASE 22 STYLES

STYLE 1:
PIN 1: EMITTER
2: BASE
3: COLLECTOR

STYLE 2:
PIN 1: SOURCE, SUBSTRATE
AND CASE
2: GATE
3: DRAIN

STYLE 3:
PIN 1: SOURCE
2: DRAIN
3: GATE

STYLE 4:
PIN 1: SOURCE
2: DRAIN
3: GATE & CASE

STYLE 5:
PIN 1: EMITTER
2: BASE 1
3: BASE 2

STYLE 6:
PIN 1: CATHODE
2: GATE
3: ANODE

STYLE 7:
PIN 1: ANODE
2: BASE
3: CATHODE

STYLE 8:
PIN 1: GATE
2: ANODE 1
3: ANODE 2

STYLE 9:
PIN 1: ANODE 2
2: ANODE 1
3: GATE
(CONNECTED TO CASE)

STYLE 10:
PIN 1: BASE
2: EMITTER
3: BASE

STYLE 11:
PIN 1: DRAIN
2: GATE
3: SOURCE, SUBSTRATE

STYLE 12:
PIN 1: SOURCE
2: GATE
3: DRAIN (CASE)

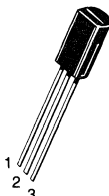
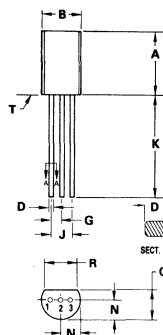


STYLE 13:
PIN 1: ANODE
2: GATE
3: CATHODE

STYLE 14:
PIN 1: ANODE
2: OPEN
3: CATHODE

CASE 29-03 (TO-226E) TO-92 1-WATT PLASTIC

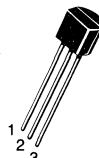
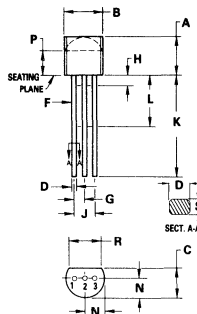
- NOTES:
1. DIMENSIONS A- AND B- ARE DATUMS.
2. -T- IS SEATING PLANE.
3. POSITIONAL TOLERANCE FOR LEADS:
 $\pm 0.10 (0.004) \text{ (T A B)}$
4. DIMENSIONING AND TOLERANCING PER
ANSI Y14.5, 1982.
5. CONTROLLING DIM: INCH.



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	7.37	7.87	0.290	0.310
B	4.44	5.21	0.175	0.205
C	3.18	4.19	0.125	0.165
D	0.46	0.61	0.018	0.024
E	1.27 BSC		0.050 BSC	
F	2.54 BSC		0.100 BSC	
G	12.70	—	0.500	—
H	2.03	2.92	0.080	0.115
J	3.43	—	0.135	—
K	0.46	0.61	0.018	0.024

CASE 29-04 (TO-226AA) TO-92 PLASTIC

- NOTES:
1. CONTOUR OF PACKAGE BEYOND ZONE "P" IS
UNCONTROLLED.
2. DIM "F" APPLIES BETWEEN "H" AND "L". DIM
"D" & "S" APPLIES BETWEEN "L" & 12.70mm
(0.51") FROM SEATING PLANE. LEAD DIM IS
UNCONTROLLED IN "H" & BEYOND 12.70mm
(0.51") FROM SEATING PLANE.
3. CONTROLLING DIM: INCH.



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	4.32	5.33	0.170	0.210
B	4.45	5.20	0.175	0.205
C	3.18	4.19	0.125	0.165
D	0.41	0.55	0.016	0.022
E	0.41	0.48	0.016	0.019
F	1.15	1.39	0.045	0.055
G	—	2.54	—	0.100
H	2.42	2.66	0.095	0.105
J	12.70	—	0.500	—
K	6.35	—	0.250	—
L	2.04	2.66	0.080	0.105
M	2.93	—	0.115	—
N	3.43	—	0.135	—
S	0.39	0.50	0.015	0.020

CASE 29 STYLES

STYLE 1:
PIN 1: EMITTER
2: BASE
3: COLLECTOR

STYLE 2:
PIN 1: BASE
2: EMITTER
3: COLLECTOR

STYLE 3:
PIN 1: ANODE
2: ANODE
3: CATHODE

STYLE 4:
PIN 1: CATHODE
2: CATHODE
3: ANODE

STYLE 5:
PIN 1: DRAIN
2: SOURCE
3: GATE

STYLE 6:
PIN 1: GATE
2: SOURCE AND SUBSTRATE
3: DRAIN

STYLE 7:
PIN 1: SOURCE
2: DRAIN
3: GATE

STYLE 8:
PIN 1: DRAIN
2: GATE
3: SOURCE AND SUBSTRATE

STYLE 9:
PIN 1: BASE 1
2: EMITTER
3: BASE 2

STYLE 10:
PIN 1: CATHODE
2: GATE
3: ANODE

STYLE 11:
PIN 1: ANODE
2: CATHODE AND ANODE
3: CATHODE

STYLE 12:
PIN 1: MAIN TERMINAL 1
2: GATE
3: MAIN TERMINAL 2

STYLE 13:
PIN 1: ANODE 1
2: GATE
3: CATHODE 2

STYLE 14:
PIN 1: EMITTER
2: COLLECTOR
3: BASE

STYLE 15:
PIN 1: ANODE 1
2: CATHODE
3: ANODE 2

STYLE 16:
PIN 1: ANODE
2: GATE
3: CATHODE

STYLE 17:
PIN 1: COLLECTOR
2: BASE
3: EMITTER

STYLE 18:
PIN 1: ANODE
2: CATHODE
3: NOT CONNECTED

STYLE 19:
PIN 1: GATE
2: ANODE
3: CATHODE

STYLE 20:
PIN 1: NOT CONNECTED
2: CATHODE
3: ANODE

STYLE 21:
PIN 1: COLLECTOR
2: EMITTER
3: BASE

STYLE 22:
PIN 1: SOURCE
2: GATE
3: DRAIN

STYLE 23:
PIN 1: GATE
2: SOURCE
3: DRAIN

STYLE 24:
PIN 1: EMITTER
2: COLLECTOR/ANODE
3: CATHODE

STYLE 25:
PIN 1: MT 1
2: GATE
3: MT 2

STYLE 26:
PIN 1: V_{CC}
2: GROUND 2
3: OUTPUT

STYLE 27:
PIN 1: MT
2: SUBSTRATE
3: MT

STYLE 28:
PIN 1: CATHODE
2: ANODE
3: GATE

STYLE 29:
PIN 1: NOT CONNECTED
2: ANODE
3: CATHODE

STYLE 30:
PIN 1: DRAIN
2: GATE
3: SOURCE

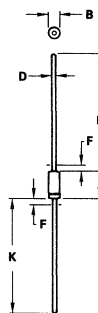
STYLE 31:
PIN 1: GATE
2: DRAIN
3: SOURCE

STYLE 32:
PIN 1: BASE
2: COLLECTOR
3: EMITTER

STYLE 33:
PIN 1: RETURN
2: INPUT
3: OUTPUT



CASE 51-02



- NOTES:
1. PACKAGE CONTOUR OPTIONAL WITHIN DIA B
AND LENGTH A. HEAT SLUGS, IF ANY, SHALL BE
INCLUDED WITHIN THIS CYLINDER, BUT SHALL
NOT BE SUBJECT TO THE MIN LIMIT OF DIA B.
2. LEAD DIA NOT CONTROLLED IN ZONES P, TO
ALLOW FOR FLASH, LEAD FINISH BUILDUP,
AND MINOR IRREGULARITIES OTHER THAN
HEAT SLUGS.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	5.84	7.62	0.230	0.300
B	2.16	2.72	0.085	0.107
D	0.46	0.56	0.018	0.022
F	—	1.27	—	0.050
K	25.40	38.10	1.000	1.500

All JEDEC dimensions and notes apply